

Mathematical Statistics

Lecture 13 The Normal Distribution Notes

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1 Lecture Overview

This lecture introduces the most important continuous probability distribution: the **Normal Distribution**, also known as the Gaussian distribution. We will explore its definition, its key properties, and most importantly, a systematic 3-step method for calculating probabilities. This distribution is the cornerstone of classical statistics and serves as a foundation for advanced topics like hypothesis testing and regression analysis.

2 The Normal Distribution and its Properties

2.1 Probability Density Function (PDF)

A continuous random variable X is said to follow a Normal distribution if its PDF is given by the function:

💡 The Normal PDF

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

for $-\infty < x < \infty$.

This distribution is defined by two parameters:

- μ (mu): The **mean** of the distribution.
- σ^2 (sigma-squared): The **variance** of the distribution (σ is the standard deviation).

We denote this relationship as: $X \sim N(\mu, \sigma^2)$.

2.2 Key Properties of the Normal Distribution

- ✔ **Bell-Shaped Curve:** The graph of the Normal PDF is a symmetric, bell-shaped curve.
- ✔ **Symmetry and Central Tendency:**
 - The curve is perfectly symmetric about its mean, μ .
 - For a Normal distribution, the **mean, median, and mode are all equal** and located at the center of the distribution.
- ✔ **Total Area is One:** As with any PDF, the total area under the curve is equal to 1.
- ✔ **Symmetrical Area:** Due to symmetry, the area to the left of the mean μ is 0.5, and the area to the right of the mean μ is also 0.5.
- ✔ **The Empirical Rule (68-95-99.7 Rule):**
 - Approximately **68%** of the data falls within 1 standard deviation of the mean ($\mu \pm 1\sigma$).
 - Approximately **95%** of the data falls within 2 standard deviations of the mean ($\mu \pm 2\sigma$).
 - Approximately **99.7%** of the data falls within 3 standard deviations of the mean ($\mu \pm 3\sigma$).

3 The Standard Normal Distribution (The Z-Distribution)

Calculating probabilities by integrating the Normal PDF directly is extremely difficult. To simplify this, we **standardize** any Normal random variable X into a **Standard Normal Variable** Z .

3.1 The Z-score Transformation

💡 The Z-score Formula

The Z-score represents how many standard deviations an observation x is away from the mean μ .

$$Z = \frac{X - \mu}{\sigma}$$

3.2 Properties of the Standard Normal Distribution

When a Normal variable $X \sim N(\mu, \sigma^2)$ is transformed, it becomes a Standard Normal variable Z with a mean of 0 and a variance of 1.

$$\text{If } X \sim N(\mu, \sigma^2), \text{ then } Z \sim N(0, 1)$$

The primary advantage is that we only need one table of probabilities—the **Standard Normal (Z) Table**—to solve problems for *any* normal distribution.

4 The 3-Step Rule for Calculating Normal Probabilities

A Systematic Approach

This simple, repeatable process can be used to solve almost any Normal distribution probability problem, turning complex calculations into a structured routine.

Step 1: Sketch, Shade, and Strategize

- Draw a bell-shaped curve.
- Mark the mean (μ) at the center.
- Mark the value(s) of X from the question on the horizontal axis.
- Shade the area corresponding to the probability you need to find.

Step 2: Convert to Z-scores

- Use the formula $Z = (X - \mu)/\sigma$ to convert your X values into Z -scores.
- This standardizes the problem to the $N(0, 1)$ distribution.

Step 3: Use the Standard Normal (Z) Table and Logic

- Look up the calculated Z -score(s) in a Standard Normal Table to find the corresponding area (probability).
- Use logic based on symmetry and the total area of 0.5 on each side of the mean to find your desired shaded region. For a table that gives area from the mean (0) to Z :
 - To find $P(Z > a)$: $0.5 - \text{Area}(0 \text{ to } a)$
 - To find $P(Z < -a)$: Same as $P(Z > a)$, by symmetry.
 - To find $P(a < Z < b)$: $\text{Area}(0 \text{ to } a) + \text{Area}(0 \text{ to } b)$ (if a and b are on opposite sides of the mean).

5 Worked Examples

5.1 Example 1: Life of Electric Bulbs

The lifetime of bulbs is normally distributed with $\mu = 800$ hours and $\sigma = 40$ hours. Find the probability that a bulb lasts:

- (a) More than 834 hours, $P(X > 834)$.
- (b) Between 778 and 834 hours, $P(778 < X < 834)$.

(a) Probability of lasting more than 834 hours

1. **Sketch & Shade:** Draw a curve centered at 800. Mark 834 and shade the tail to the right.
2. **Convert to Z-score:** $Z = (834 - 800)/40 = 34/40 = 0.85$. The problem becomes $P(Z > 0.85)$.
3. **Use the Z-Table:** A standard table gives the area from the mean to Z . $\text{Area}(0 \text{ to } 0.85)$ is typically around 0.3023. The total area right of the mean is 0.5.

$$P(Z > 0.85) = 0.5 - \text{Area}(0 \text{ to } 0.85) = 0.5 - 0.3023 = \mathbf{0.1977}$$

(b) Probability between 778 and 834 hours

1. **Sketch & Shade:** Draw a curve centered at 800. Mark 778 (left) and 834 (right). Shade the area between them.

2. **Convert to Z-scores:**

$$Z_1 = (778 - 800)/40 = -22/40 = -0.55$$

$$Z_2 = (834 - 800)/40 = 34/40 = 0.85$$

The problem becomes $P(-0.55 < Z < 0.85)$.

3. **Use the Z-Table:** The area is Area(-0.55 to 0) + Area(0 to 0.85). By symmetry, Area(-0.55 to 0) = Area(0 to 0.55). From tables: Area(0 to 0.55) $\approx$ 0.2088 and Area(0 to 0.85) $\approx$ 0.3023.

$$\text{Total Probability} = 0.2088 + 0.3023 = \mathbf{0.5111}$$

5.2 Example 2: The Reverse Problem

For a normally distributed variable, 31% of items are under 45, and 8% are over 64. Find μ and σ .

1. **Translate and Sketch:**

- $P(X < 45) = 0.31 \implies$ 45 is left of the mean. Area(45 to μ) = $0.5 - 0.31 = 0.19$.
- $P(X > 64) = 0.08 \implies$ 64 is right of the mean. Area(μ to 64) = $0.5 - 0.08 = 0.42$.

2. **Reverse Z-lookup:**

- The Z-score whose area from 0 is 0.19 is $Z_1 \approx -0.50$ (negative because it's on the left).
- The Z-score whose area from 0 is 0.42 is $Z_2 \approx 1.40$.

3. **Solve System of Equations:**

$$-0.50 = (45 - \mu)/\sigma \tag{1}$$

$$1.40 = (64 - \mu)/\sigma \tag{2}$$

Subtracting (1) from (2): $1.90\sigma = 19 \implies \sigma = \mathbf{10}$.

Substitute into (1): $-0.50(10) = 45 - \mu \implies -5 = 45 - \mu \implies \mu = \mathbf{50}$.

Conclusion: The mean is 50 and the standard deviation is 10.

6 Practical Applications in Data Science & ML

Applications of the Normal Distribution

- ⚙️ **Hypothesis Testing:** The normal distribution is the foundation for Z-tests and T-tests, which are used to determine if results from experiments (like A/B testing a website change) are statistically significant or just due to random chance.
- ⚙️ **Confidence Intervals:** When estimating a population parameter (e.g., average ad click-through rate), we construct confidence intervals using the normal distribution to quantify the range in which the true value likely lies.
- ⚙️ **Model Assumption Checking:** In linear regression, a key assumption is that the errors (residuals) are normally distributed. Data scientists plot a histogram or a Q-Q plot of residuals to validate this assumption.
- ⚙️ **Central Limit Theorem (CLT):** The CLT states that the distribution of sample means will be approximately normal, regardless of the original population's distribution, given a large enough sample size. This theorem is why the normal distribution appears so frequently in statistics.

7 The Core Concept: The 3-Step Solution Framework

From Theory to Practice

This fundamental workflow for solving Normal Distribution problems bridges the gap between abstract formulas and concrete, real-world answers.

- 1. Define the Random Variable and Parameters:** Identify what X represents (e.g., sales, weight), and note the given mean (μ) and standard deviation (σ).
- 2. Translate to a Probability Statement:** Convert the problem's objective into a mathematical statement, such as $P(X > 40000)$ or $P(30000 < X < 40000)$.
- 3. Apply the 3-Step Calculation Method:**

Step A: Sketch: Draw a bell curve, mark the mean μ and the X value(s), and shade the desired area. This visual guide is crucial for avoiding errors.

Step B: Standardize: Convert all X values into Z-scores using the formula:

The Z-Score Formula

$$Z = \frac{X - \mu}{\sigma}$$

Step C: Solve: Use a Standard Normal (Z) table and the properties of the distribution (symmetry, total area = 1) to find the probability of the shaded area.

8 Example 1: Analyzing Business Sales

8.1 Problem Statement

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A sales tax officer reports that for a group of 500 businesses, the average sales (μ) are ₹36,000 with a standard deviation (σ) of ₹10,000. Assuming sales are normally

12 Lecture Summary

📅 Key Takeaways

- 💡 The **Normal Distribution** is a symmetric, bell-shaped curve defined by its mean (μ) and variance (σ^2).
- 💡 We use the **Z-score** $Z = (X - \mu)/\sigma$ to standardize any normal variable into the **Standard Normal Distribution** $N(0, 1)$.
- 💡 The **3-Step Rule** provides a robust method for solving probability problems:
 1. **Sketch and Shade** the required area.
 2. **Convert to Z-scores** to standardize.
 3. **Use the Z-Table and Logic** to find the area.
- 💡 The Normal distribution is the backbone of inferential statistics and is widely used in data science applications.

13 Practice Problems

1. The weights of bags of cement are normally distributed with a mean of 50 kg and a standard deviation of 2 kg.
 - (a) What is the probability that a randomly selected bag weighs less than 47 kg?
 - (b) What is the probability that a bag weighs between 49 kg and 53 kg?
2. The scores on a standardized test are normally distributed with a mean of 100 and a standard deviation of 15. The top 10% of students receive a special commendation. What is the minimum score required to receive it?
3. A machine produces bolts with a diameter that is normally distributed with a mean of 10 mm and a standard deviation of 0.1 mm. A bolt is considered defective if its diameter is less than 9.8 mm or greater than 10.2 mm. What is the percentage of defective bolts produced?
4. The daily return of a stock is approximately normally distributed with a mean of 0.05% and a standard deviation of 1.5%. What is the probability that the stock will lose money (have a return less than 0%) on a given day?